

# CHAPTER 2

# NUTRITION IN PREGNANCY, LACTATION AND INFANCY

## OBJECTIVES

**This chapter briefly discuss about:-**

- Physiological changes during pregnancy
- Nutritional requirments of pregnant and lactating women, consequences of malnutrition during pregnancy and lactation.
- Nutritional needs of infants, advantages of breast milk, artificial feeding, weaning and problems of weaning.

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Pregnancy and lactation are the most stressful periods in the life of a woman. Adequate nutrition before and during pregnancy has greater potential for a long term health impact than it does at any other time. Studies in human and animals indicate that nutrition during pregnancy influences not only the normal development of the foetus and immediate help of the new-born infants, but there are now also compelling data that nutrition during pregnancy influences the subsequent morbidity and mortality of the off - spring when they are grown adults. Current recommendation for the nutrition during pregnancy stress the importance of a proper pattern of weight gain and an adequate intake of calories, proteins, vitamins and minerals to allow for optimal foetal development and the preservation of maternal health.

## 2.1 Nutrition in pregnancy

A woman who has been well nourished before conception begins her pregnancy with reserves of several nutrients so that the needs of the growing foetus can be made without affecting her health.

### 2.1 a. Physiological changes during pregnancy

- Total plasma volume in a non - pregnant women averages 2600 ml. By 34 weeks it is about 50% greater than it was at conception.
- Increased blood volume produces a high glomerular filtration rate. It appears that the renal tubules are unable to adjust completely and a percentage of nutrients that would have been reabsorbed in the non - pregnant women are excreted in the urine.



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